

Auditing Hidden Routing Regimes in Black-Box Models

Consolidated method note (v3): definition, inputs/outputs, assumptions, algorithm,
and a controlled evaluation with known regimes

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Abstract

This note consolidates the v2 algorithm note (*routing_audit_v2*, July 24) and its derivation guide into a single description of the *current* method, incorporating the experimental findings of July 28 that changed the algorithm substantially: modality (the dip test), not dispersion, is the primary detection statistic; the hierarchical GP null generator is no longer required; the probe family is decisive rather than a reporting stratification; query allocation is a first-class design parameter; and local $K=2$ recovery of the routing penalty is solved while general- K global synchronization remains open. The note gives, in order: the definition of a regime and the recoverable estimand (§1); the method’s inputs and outputs (§2); the complete assumption set, including two new preconditions surfaced by experiment (§3); the complete algorithmic steps (§4); the evidence that forced each change from v2 (§5); a new controlled simulation with known true regimes evaluating recovery under a $40\times$ noise sweep and near regime boundaries (§6); and a like-for-like comparison against the previous graph-based (SBM) approach and three simple baselines (§7). Status and open problems close the note (§8).

1 Setting and the definition of a regime

A deployed black-box system is modeled as

$$f(x) = f_{g(x)}(x) + \eta,$$

where $g : \mathcal{X} \rightarrow [K]$ is an unobserved routing rule, f_k is the response mechanism of branch k , and η is observation noise. **A regime is a pair:** a branch f_k together with the region $g^{-1}(k)$ on which the router selects it. The auditor observes only queries and responses through a response map ψ ; in the present applications the response is scalar ($V = 1$), which collapses the vector machinery of v2 (whitening is division by a scalar; the dip is applied to the responses directly, with no projection choice).

Identifiability ceiling. Regimes are recoverable only up to behavioural equivalence: two branches that answer identically under ψ at the probed inputs are indistinguishable in principle, and a sharp-but-continuous transition narrower than the probed scales is observationally equivalent to a hard gate. All claims are therefore relative to ψ , the probe design, and the declared smoothness class. (The controlled evaluation in §6 sharpens this: the dip statistic responds to *jumps* and not to continuous kinks, so the observational-equivalence class is narrower than v2 assumed.)

The estimand is the boundary, not the partition. Detection has a radius: a gate is visible only from anchors whose probe neighbourhood actually straddles it. The natural estimand is therefore the boundary — the level set where g flips — together with boundary-adjacent territory; per-anchor regime classification is a derived quantity, and “unflagged” is never read as “interior.” The corrected ground-truth criterion used throughout: an anchor is *detectable* at scale σ when the minority probe mass across the gate, $\pi_{\text{true}}(\sigma)$, is at least 0.05; truth is mixing fraction, not distance.

2 Inputs and outputs

Inputs. (i) query access to f (no labels, no internals, no gradients); (ii) a set of anchors $x_1, \dots, x_n$ (real instances); (iii) a probe family and a scale ladder $\sigma_1 < \dots < \sigma_T$ — *on-manifold* (kernel-weighted resampling of real inputs) as primary where the covariate distribution is locally smooth (assumption A12), ambient Gaussian otherwise or as robustness check; (iv) a response map ψ ; (v) a per-anchor query budget m per scale, with the allocation rule of §4. No protected attribute enters the audit; it is used only in Stage C scoring.

Outputs. Per anchor: a flag with a p -value (dip, Bonferroni over the scales actually dipped); the recovered penalty $\hat{\Delta}$ (component-mean gap of a $K=2$ mixture at the most significant scale) with a consistency measure across scales; the soft local composition $\hat{\pi}$; the anchor orientation $\hat{z}$ (which branch owns the unperturbed response); and an *abstention* when support or signal is insufficient — a first-class output, distinct from “null-consistent.” Audit level: an alignment claim — do flags concentrate near candidate gate boundaries? — tested by spatial (or graph) randomization with radius-matched statistics; abstention and negative-control rates; and, when per-anchor confirmatory claims are needed, Benjamini–Yekutieli across anchors as a dependence-robust sensitivity.

3 Assumptions

B1–B10 carry over from v2 §8 unchanged in content; stated compactly with diagnostics:

- **B1 stationarity.** One routing law throughout the audit (no stateful adversary). *Diag:* interleaved disguised queries; batch drift tests.
- **B2 probe accessibility.** Recovered regimes are those some probe reaches; $\widehat{K}$ counts observable regimes. *Diag:* cross-support comparison.
- **B3 local distinguishability.** Locally present regimes differ in response law with bounded local classification error. *Diag:* held-out likelihood, stability.
- **B4 prevalence.** Minor components need soft mass $m\hat{\pi} \geq n_{\min}$ ($n_{\min} = 20$ here). *Diag:* report soft masses; merge or omit failures.
- **B5 within-regime continuity.** Branch response laws drift Lipschitz-slowly between neighbouring anchors. *Diag:* centroid smoothness along the anchor graph.
- **B6 matching margin** and **B7 overlap connectivity** — required only for general- K synchronization (open; §8); trivial at $K=2$, where components are ordered by mean.
- **B8 anchor fidelity.** The unperturbed query is routed by the same mechanism (used only for orientation). *Diag:* repeated unperturbed queries.

- **B9 gate regularity** and **B10 transversality**. Locally linear boundary at probe scale; probes must have variance along the boundary normal. *Diag*: held-out label prediction; support comparison.

Two preconditions the July experiments added, both *checkable*:

- **A11 (honest-model smoothness at every probed scale)**. The single-branch surface must be effectively straight at all ladder scales: lengthscale $\ell > \sigma_T$ and smooth amplitude far below the gap of interest. Verified for Seattle two independent ways: $\ell \approx 2367$ m against a ladder top of 1200 m, and local curvature amplitude $s \approx 0.004$ log-EUI against a penalty of 0.30 (75 $\times$). The simulation (§6) shows the condition is *sharp*: GP nulls with ℓ above the ladder top flag at 0.00–0.04; with ℓ below it, at 0.33–0.81. Operationally: *cap the ladder below the honest lengthscale*, or declare the smoothness class — this is the surviving content of v2’s $\ell_{\min}$ registration, now attached to the dip rather than to a null generator.
- **A12 (covariate local smoothness, for on-manifold probes)**. The probe distribution itself must be locally smooth in the probe metric; otherwise the data’s own clustering is read as model multimodality. This is what defeated the attribute-graph experiment (construction-boom years, standard floor counts: dip $p \approx 0$ on the raw covariates). Checkable by dipping the covariates; ambient probes satisfy it by construction, which partially offsets their off-manifold disadvantage. Geography satisfied it by luck, not principle.

4 The algorithm

The simplified pipeline. Stage A’ makes the existence and localization claims; Stage B’ recovers local structure at $K=2$; Stage C’ makes the audit-level claims. The dispersion apparatus of v2 §3.1–3.4 and the hierarchical GP null generator of v2 §3.3 are *not part of the pipeline* (see §5 for the evidence).

Stage A’: detection by naive multi-scale dip scan.

- A1. Register** the design: ψ ; probe families; a ladder of 3–6 log-spaced scales capped by A11; per-scale budget m ; α ; the effect-size threshold; negative controls; support diagnostics.
- A2. Probe** each anchor at each scale: on-manifold primary (kernel-weighted resample of real inputs, weights $\propto e^{-d^2/2\sigma^2}$), abstaining at scales with fewer than 12 distinct neighbours; ambient as robustness family.
- A3. Minimum-signal rule, per scale**: dip a scale only if the response variance clears the observation-noise floor (χ^2 test at 5%). Scales failing it are abstained from, not counted as evidence of absence.

Why: applied per scale, not per anchor; the first Seattle run produced $R^{\log} \approx 18$ purely from an epsilon clamp when this was done per anchor.

- A4. Dip** the (whitened) responses at each surviving scale; **flag** the anchor if $\min_t p_t < \alpha/\#\{\text{scales dipped}\}$ (Bonferroni across scales). No dispersion curve, no σ^* .
- A5. Query allocation**. Power is limited by the minor-mode mass $m\pi$, not by the test. Prefer fewer scales with larger m : on the weak band $\pi_{\text{true}} \in [0.05, 0.10]$ at equal budget, $3 \times m=1000$ gives power 0.875 versus 0.472 for $6 \times m=500$. Design rule: keep $m\pi$ above the dip’s power threshold in the weakest band that matters.

Stage B': local $K=2$ recovery.

- B1. Fresh draw** per scale (budget split respected: the recovery sample is never the detection sample).
- B2. Mixture.** Fit a 2-component Gaussian mixture; order components by mean; retain soft responsibilities. Report $\hat{\pi}$ (mean responsibility), and $\hat{\Delta}$ = component-mean gap *at the scale of minimum dip p -value* (the across-scale median dilutes $\hat{\Delta}$ with non-crossing scales), plus its cross-scale consistency (CV). Apply the minimum-mass rule $m\hat{\pi} \geq 20$.
- B3. Effect-size filter.** Keep flags with $\hat{\Delta}$ above a registered threshold (Seattle: $\hat{\Delta} > 0.15$ lifted precision $0.968 \rightarrow 1.000$ at 652 flags). Report as a precision/recall curve, not a single point: $\hat{\Delta}$ is biased downward at small π , so a fixed threshold trades recall in the weak band.
Why: $\hat{\Delta}$ and the dip read the same bimodality — this is one test plus a practical-significance filter, not independent corroboration.
- B4. Orientation.** Posterior-assign the unperturbed response to a component; at $K=2$ synchronization is ordering by mean. General K : open (v2 SYNC1–7 remains the committed procedure; see §8).

Stage C': audit-level claims.

- C1. Alignment by randomization.** Rather than 3,581 dependent per-anchor claims, test what the audit cares about: do flags concentrate near candidate boundaries? Permute the gating covariate across units (tracts), holding geometry and anchors fixed — valid under arbitrary spatial dependence, no null generator. **Radius-match** the test statistic to the method's measured reach and report the full radius sweep (Seattle, on-manifold, ≤ 800 m: $p = 0.0025$; a mismatched ≤ 100 m statistic penalizes the method's own detection radius).
- C2. Per-anchor claims** (if needed): BY at $q = 0.10$ across anchors as the dependence-robust correction (Seattle flags survive the $c(n) = 8.68$ penalty).
- C3. Harm read-out** (optional): group-conditional conformal coverage. Seattle: marginal coverage 0.898 at nominal 90%, but 0.840 on the penalized branch versus 0.903 off it — invisible marginally, and the recovered partition is exactly the grouping a Mondrian conformal predictor needs.
- C4. Report** existence, structure, interpretation separately, with abstentions, controls, and certified-versus-extrapolated territory as in v2.

5 What changed from v2, and the evidence that forced it

- **Dispersion demoted from primary statistic to (at most) optional corroborator.** It fails all three roles tested: not a detector (uncalibrated $R^{\log}$ flags smooth GP nulls at 0.65–1.00; calibrated T^{scale} reaches power 0.025 or 0.000 depending on the null's construction); not a scale selector (σ^* is *worse than a random scale*: power 0.715 vs 0.798, because $r(\sigma)$ peaks small while the dip wants balanced mixing); not an efficient screen (AUC 0.750; retaining 90% of flags still requires dipping 70% of anchors, at $\sim 4\times$ the queries it saves). Its surviving corroboration value (precision $0.52 \rightarrow 0.81$, $\kappa = 0.32$) is dominated by the $\hat{\Delta}$ filter.

- **Modality promoted to primary.** A hard gate seen through a probe of scale σ is observationally a transition of width $\approx \sigma$, and a GP with $\ell \approx \sigma$ reproduces the same dispersion curve — *no statistic computed from $r(\sigma)$ alone can separate them*. What differs is the response *distribution*: a gate makes two clumps separated by the penalty. Seattle: dip flag rate 0.213 gated vs 0.000 honest across 3,420 buildings.
- **Null generator dropped.** It existed to calibrate dispersion; with dispersion gone, the $\ell_{\min}$ registration problem, the contamination/cross-fitting machinery, and the misspecification exposure go with it. The surviving smoothness declaration is A11, attached directly to the ladder.
- **Probe family is decisive, not a caveat.** The spatial alignment claim is significant on-manifold ($p = 0.013$ – 0.033 across all four statistics) and never ambient ($p = 0.083$ – 0.208): long-range ambient flags dilute the alignment. On-manifold is primary *where A12 holds* — the attribute-graph negative result is precisely an A12 violation, and ambient probes are the fallback there.
- **Estimand: boundary-adjacent territory** (detection radius $\approx$ probe reach; §6 table by distance), with per-anchor labels derived.
- **Query allocation first-class** (A5 above): reallocation lifts projected Seattle power on detectable anchors from 0.737 to ~ 0.94 .
- **Stage B split.** Local $K=2$ recovery with a pure anchor is *solved* (Seattle: $\hat{\Delta} = 0.291$ against a planted 0.300; honest-model placebo 0.014). General- K synchronization, certification, and the recovery theorem remain open.

6 Controlled evaluation with known regimes

New in this consolidation (code and outputs in `sim/`; settings registered in file headers; seed 20260729). Two constructions, both with the audit blind to ground truth: **1-D**: $f(x) = \beta x - \Delta \mathbf{1}[x \geq 0.5]$, $\beta = 0.15$, $\Delta = 0.30$, ladder $\{0.02, 0.063, 0.2\}$, $m_{\text{dip}} = m_{\text{rec}} = 1000$; matched no-gate control; 240 anchors. **2-D**: wavy boundary $x_2 = 0.5 + 0.08 \sin 4\pi x_1$, honest model linear plus a long-lengthscale GP term ($\ell = 0.5 > \sigma_T = 0.2$; amplitude $15\times$ below Δ), 480 anchors — the Seattle geometry class. Both satisfy the measured Seattle ordering (A11), and both report the corrected ground truth π_{true} .

Noise sweep (Peng’s first axis). τ_{obs} swept $0.005 \rightarrow 0.2$ (Δ/τ from $60\times$ to $1.5\times$):

τ_{obs}	0.005	0.01	0.02	0.05	0.10	0.20
power (detectable anchors)	0.976	0.964	0.892	0.567	0.012	0.000
FP, matched no-gate model	0.000	0.000	0.000	0.000	0.000	0.004
abstention, no-gate model	0.000	0.000	0.000	0.000	0.325	0.792
$\hat{\Delta}$ at $p_{\min}$ scale (true 0.30)	0.248	0.248	0.248	0.252	0.288	—

False positives stay at or below 0.004 across the entire sweep; when noise drowns the signal the method *abstains* rather than flags (the per-scale minimum-signal rule); and wherever anything is flagged, the recovered penalty is within 0.05 of truth with a mild, explainable downward bias (the mixture absorbs trend width into the components).

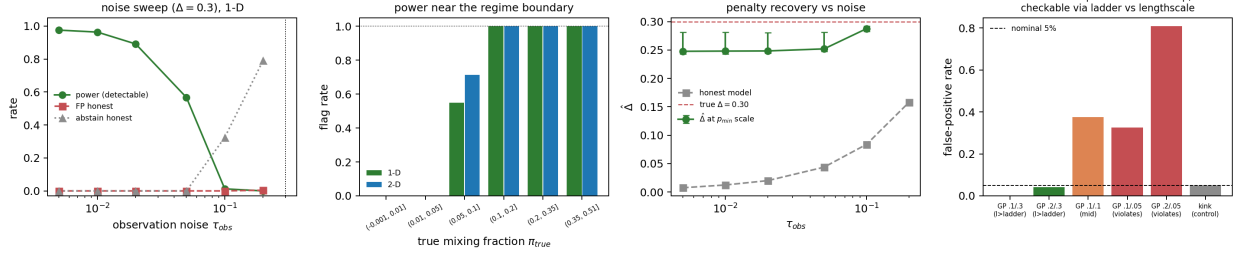


Figure 1: Known-regimes simulation. Left to right: noise sweep (power, FP, abstention); power against true mixing fraction near the boundary (1-D and 2-D); penalty recovery against noise with the honest-model placebo; the A11 precondition is sharp and checkable.

Near regime boundaries (Peng’s second axis). Power against the true mixing fraction, the controlled analogue of the Seattle power-vs- π curve:

π_{true}	$\leq .01$	.01–.05	.05–.10	.10–.20	.20–.35	.35–.50
flag rate, 1-D	0.00	0.00	0.55	1.00	1.00	1.00
flag rate, 2-D	0.00	0.00	0.71	1.00	1.00	1.00

Detection is binary in π : perfect above 0.10, impossible below 0.05 ($m\pi < 50$ minor-mode points), partial in between — confirming that power is limited by minor-mode mass, not by the test, and motivating the allocation rule. By distance (2-D): flag rate 1.00 out to the ladder top, 0.73 at 1–1.5 $\times$, 0.015 beyond — the detection radius *is* probe reach. Orientation accuracy on flagged anchors: 0.956; overall 2-D: power 0.929, FP 0.000, $\hat{\Delta}$ median 0.270 (0.297 within 0.02 of the boundary).

Robustness, and a sharp new precondition (A11). Smooth GP confounders and a kink control at $\tau = 0.02$:

null model	ℓ vs ladder top	FP rate	reading
GP $A=.10$, $\ell=.30$	above	0.000	precondition holds
GP $A=.20$, $\ell=.30$	above	0.042	precondition holds
GP $A=.10$, $\ell=.10$	straddles	0.375	marginal
GP $A=.10$, $\ell=.05$	below	0.325	violated — designed failure
GP $A=.20$, $\ell=.05$	below	0.808	violated — designed failure
kink (continuous, unrouted)	—	0.050	<i>not</i> flagged; dip targets jumps

Dipping at scales beyond the honest lengthscale reads smooth wiggles as modes; capping the ladder (checkable; Seattle had a factor of two to spare) restores nominal error. The kink control refines v2’s observational-equivalence discussion: the dip does *not* flag sharp-but-continuous behaviour at nominal rates — modality is specific to jumps, a specificity dispersion never had.

7 Comparison with the graph-based approach and simple baselines

All methods run on the same 2-D gated/honest pair. The SBM row runs the April behavioural-fingerprint thread’s own code (`geospatial-xai-attacks/src/sbm`: kNN cosine graph over per-anchor response fingerprints, collapsed-Gibbs Bernoulli SBM, $K=2$, modal partition).

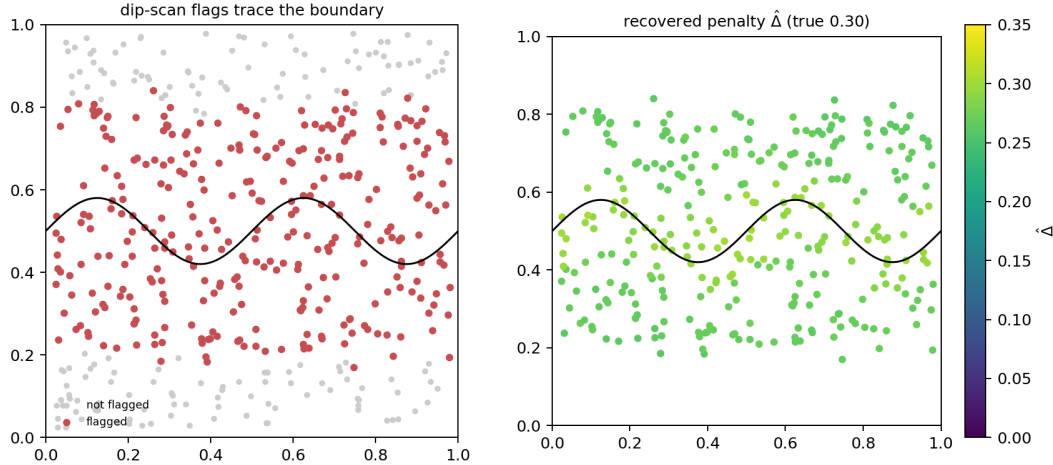


Figure 2: 2-D spatial gate: flags trace the wavy boundary (left); recovered $\hat{\Delta}$ per flagged anchor against a planted 0.30 (right).

method	existence FP (honest)	partition acc. (all anchors)	partition acc. (detectable)	$\hat{\Delta}$ err	queries /anchor
dip-scan (ours)	0.000	0.80	0.93	0.03	6001
global cluster + BIC	0.00	0.54	0.54	0.30	25
residual outlier	0.008	0.52	0.52	—	25
SPA profile-mean	<i>no test</i>	0.52	0.58	0.21	50
plain SBM, $K=2$ (their code)	<i>no test</i>	0.61	—	—	33

Readings. **Global clustering** answers *existence* in this easy setting (BIC picks $K=2$ gated, $K=1$ honest) but its partition is chance-level: the detrending fit absorbs most of the gate, and the residual split follows covariates rather than mechanism — the failure v2 §5 predicted. **Residual outliers** see almost nothing (the smooth fit absorbs the level shift). **SPA** has no calibrated existence claim and near-chance partition; its vertex gap over-estimates from noise extremes ($\hat{\Delta}$ error 0.21) — averaging destroys the modality evidence, which was the original reason for abandoning the route. **The SBM** aligns only weakly with the true regime (ARI 0.048 gated, 0.000 honest) and returns a partition unconditionally: behavioural-fingerprint similarity cancels the constant penalty (a level effect) except across boundaries, so the block structure has little to latch onto — intrinsic to similarity-graph methods, not a tuning failure. Ours is the only method with a calibrated per-anchor existence claim, boundary localization, and penalty recovery; its cost, $\sim 200\times$ the query budget of the cheap baselines, is the price of those guarantees and is the axis the allocation rule optimizes.

8 Status and open problems

Demonstrated (Seattle, $n = 3,581$, planted gate 0.30 on tract share, and the controlled suite above): existence with 0.0075 honest flag rate and power 0.737 ($\rightarrow \sim 0.94$ reallocated); precision 1.000 at the registered operating point ($\hat{\Delta} > 0.15$, 652 flags); recovery $\hat{\Delta} = 0.291$ (IQR [0.152, 0.299]) vs placebo 0.014; alignment $p = 0.0025$ radius-matched; conditional-coverage harm invisible to a marginal conformal check.

Open, in rough order of value:

1. *General-K synchronization and certification* (v2 SYNC1–7, CERT1–4) with the theorem-shaped target of v2 §13: forest propagation correct with probability $\geq 1 - \sum_e \alpha_e$ under B3–B7; unblocked but unstated.
2. *Generalization beyond geospatial*: the anchor graph need not be geographic (v2 SYNC1), but A12 must hold in the probe metric. Routes: discrete graphs (feeder, rate class, tariff), proper residualization (the null generator’s job resurfacing in a smaller role — current partial fix reduces honest FP $0.27 \rightarrow 0.10$, still $2\times$ nominal), or a smoother dataset.
3. $\hat{\Delta}$ at small π : downward bias (0.158 on the Seattle weak band); report filter as precision/recall curve; pooling across the anchor graph is the likely fix and is the same machinery as (1).
4. *Adaptive adversaries* (response smoothing near probe scales) — scoped in v2, untouched.

Reproducibility. The simulation suite is self-contained (numpy/pandas/ matplotlib; optional scipy/diptest/sklearn), seeded, and resumable; the Seattle pipeline’s reorganization into `geospatial-xai-attacks` with configs and a single `run_all` entry point accompanies this note.